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DAY 19 /100

JAVA INTERVIEW - CODING ROUND

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01 Full Pyramid

```
    *
   ***
  *****
 *******
*********
```

```java
/**
 * Solution for Full Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops to manage spaces and stars
 */
public class FullPyramid
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n; i++)
        {
            for (int j = i; j < n; j++)
            {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2 * i - 1); j++)
            {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

```java
/**
 * Solution for Full Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using String.repeat() to simplify printing
 */
public class FullPyramid
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n; i++)
        {
            System.out.print(" ".repeat(n - i));
            System.out.println("*".repeat(2 * i - 1));
        }
    }
}
```

02 Right Half Pyramid

```
*
* *
* * *
* * * *
* * * * *
```

```java
/**
 * Solution for Right Half Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops to manage rows and stars
 */
public class RightHalfPyramid
{
    public static void main(String[] args)
    {
        int n = 5;
        for (int i = 1; i <= n; i++)
        {
            for (int j = 1; j <= i; j++)
            {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

```java
/**
 * Solution for Right Half Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using String.repeat() to simplify star printing
 */
public class RightHalfPyramid
{
    public static void main(String[] args)
    {
        int n = 5;
        for (int i = 1; i <= n; i++)
        {
            System.out.println("* ".repeat(i));
        }
    }
}
```

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03 Diamond Pattern

```
    *
   * *
  * * *
   * *
    *
```

```java
/**
 * Solution for Diamond Pattern
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops for upper and lower parts of the diamond
 */
public class DiamondPattern
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n; i++)
        {
            System.out.print(" ".repeat(n - i));
            System.out.println("* ".repeat(i));
        }
        for (int i = n - 1; i >= 1; i--)
        {
            System.out.print(" ".repeat(n - i));
            System.out.println("* ".repeat(i));
        }
    }
}
```

```java
/**
 * Solution for Diamond Pattern
 * Author: Java Tech Community by PKC
 * Approach: Combining both parts of the diamond in a single loop
 */
public class DiamondPattern
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n * 2   1; i++)
        {
            int spaces = Math.abs(n - i);
            int stars = n   spaces;
            System.out.print(" ".repeat(spaces));
            System.out.println("* ".repeat(stars));
        }
    }
}
```

04 Hourglass Pattern

```
* * * * *
 * * * *
  * * *
   * *
    *
   * *
  * * *
 * * * *
* * * * *
```

```java
/**
 * Solution for Hourglass Pattern
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops for the hourglass structure
 */
public class HourglassPattern
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = n; i >= 1; i--)
        {
            System.out.print(" ".repeat(n - i));
            System.out.println("* ".repeat(i));
        }
        for (int i = 2; i <= n; i++)
        {
            System.out.print(" ".repeat(n - i));
            System.out.println("* ".repeat(i));
        }
    }
}
```

```java
/**
 * Solution for Hourglass Pattern
 * Author: Java Tech Community by PKC
 * Approach: Combining top and bottom parts into a single loop
 */
public class HourglassPattern
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n * 2 - 1; i++)
        {
            int stars = (i <= n) ? (n - i + 1) : (i - n + 1);
            int spaces = Math.abs(n   i);
            System.out.print(" ".repeat(spaces));
            System.out.println("* ".repeat(stars));
        }
    }
}
```

05 Hollow Full Pyramid

```
    *
   * *
  *   *
 *     *
* * * * *
```

```java
/**
 * Solution for Hollow Full Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops with conditions to create hollow spaces
 */
public class HollowFullPyramid
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n; i++)
        {
            for (int j = i; j < n; j++)
            {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2 * i - 1); j++)
            {
                if (j == 1 || j == (2 * i - 1) || i == n)
                {
                    System.out.print("*");
                }
                else
                {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

```java
/**
 * Solution for Hollow Full Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using String.repeat() and conditions for hollow spaces
 */
public class HollowFullPyramid
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n; i++)
        {
            System.out.print(" ".repeat(n   i));
            for (int j = 1; j <= (2 * i - 1); j++)
            {
                if (j == 1 || j == (2 * i - 1) || i == n)
                {
                    System.out.print("*");
                }
                else
                {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

06 Hollow Inverted Full Pyramid

```
* * * * *
 *     *
  *   *
   * *
    *
```

```java
/**
 * Solution for Hollow Inverted Full Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops with conditions to manage hollow spaces
 */
public class HollowInvertedFullPyramid
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = n; i >= 1; i--)
        {
            for (int j = n; j > i; j--)
            {
                System.out.print(" ");
            }
            for (int j = 1; j <= (2 * i - 1); j++)
            {
                if (j == 1 || j == (2 * i - 1) || i == n)
                {
                    System.out.print("*");
                }
                else
                {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

```java
/**
 * Solution for Hollow Inverted Full Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using String.repeat() with conditions for hollow spaces
 */
public class HollowInvertedFullPyramid
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = n; i >= 1; i--)
        {
            System.out.print(" ".repeat(n - i));
            for (int j = 1; j <= (2 * i - 1); j++)
            {
                if (j == 1 || j == (2 * i - 1) || i == 1)
                {
                    System.out.print("*");
                }
                else
                {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

07 Hollow Diamond Pyramid

```
    *
  *   *
*       *
  *   *
    *
```

```
 **
 * Solution for Hollow Diamond Pyramid
   Author: Java Tech Community by PKC
 * Approach: Using nested loops for top and bottom halves with hollow spaces
 */
public class HollowDiamondPyramid
{
                  void main(String[] args)

        int n = 5; // Number of rows
        for (int i = 1; i <= n; i++)
        {
            System.out.print(" ".repeat(n - i));
            for (int j = 1; j <= (2 * i - 1); j++)
            {
                            | j == (2 * i - 1))

                    System.out.print("*");
                }
                else
                {
                    System.out.print(" ");

            }
            System.out.println();
        }
        for (int i = n - 1; i >= 1; i--)
        {
            System.out.print(" ".repeat(n - i));
            for (int j = 1; j <= (2 * i - 1); j++)
            {
                if (j == 1 || j == (2 * i - 1))
                {
                    System.out.print("*");
                }
                else
                {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
```

```
/**
 * Solution for Hollow Diamond Pyramid
 * Author: Java Tech Community by PKC
 * Approach: Using a single loop for both halves with space and star conditions
 */
public class HollowDiamondPyramid
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 1; i <= n * 2 - 1; i++)
        {
            int spaces = Math.abs(n - i);
            int stars = (n   spaces) * 2 - 1;
            System.out.print(" ".repeat(spaces));
            for (int j = 1; j <= stars; j++)
            {
                if (j == 1 || j == stars)
                {
                    System.out.print("*");
                }
                else
                {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

08 Floyd’s Triangle

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

```java
/**
 * Solution for Floyd’s Triangle
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops to print consecutive numbers row-wise
 */
public class FloydsTriangle
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        int num = 1; // Starting number
        for (int i = 1; i <= n; i++)
        {
            for (int j = 1; j <= i; j++)
            {
                System.out.print(num++ + " ");
            }
            System.out.println();
        }
    }
}
```

```java
/**
 * Solution for Floyd’s Triangle
 * Author: Java Tech Community by PKC
 * Approach: Using a single loop and keeping track of row limits
 */
public class FloydsTriangle
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        int num = 1;
        for (int i = 1; i <= n; i++)
        {
            System.out.println(java.util.stream.IntStream.range(num, num + i)
                                .mapToObj(String::valueOf)
                                .collect(java.util.stream.Collectors.joining(" ")));
            num += i;
        }
    }
}
```

09 Pascal's Triangle

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

```java
/**
 * Solution for Pascal's Triangle
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops with combination formula
 */
public class PascalsTriangle
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        for (int i = 0; i < n; i++)
        {
            System.out.print(" ".repeat(n - i));
            int value = 1;
            for (int j = 0; j <= i; j++)
            {
                System.out.print(value + " ");
                value = value * (i - j) / (j + 1);
            }
            System.out.println();
        }
    }
}
```

```java
/**
 * Solution for Pascal's Triangle
 * Author: Java Tech Community by PKC
 * Approach: Using a 2D array to store values for each row
 */
public class PascalsTriangle
{
    public static void main(String[] args)
    {
        int n = 5; // Number of rows
        int[][] pascal = new int[n][n];

        for (int i = 0; i < n; i++)
        {
            // Print leading spaces
            System.out.print(" ".repeat(n - i));
            for (int j = 0; j <= i; j++)
            {
                // Calculate Pascal value
                if (j == 0 || j == i)
                {
                    pascal[i][j] = 1;
                }
                else
                {
                    pascal[i][j] = pascal[i - 1][j - 1] + pascal[i - 1][j];
                }
                System.out.print(pascal[i][j] + " ");
            }
            System.out.println(); // Move to the next row
        }
    }
}
```

10 Hollow Square Pattern

```
* * * * *
*       *
*       *
*       *
* * * * *
```

```java
/**
 * Solution for Hollow Square Pattern
 * Author: Java Tech Community by PKC
 * Approach: Using nested loops with conditions for borders
 */
public class HollowSquarePattern
{
    public static void main(String[] args)
    {
        int n = 5; // Size of the square
        for (int i = 1; i <= n; i++)
        {
            for (int j = 1; j <= n; j++)
            {
                if (i == 1 || i == n || j == 1 || j == n)
                {
                    System.out.print("* ");
                }
                else
                {
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

```java
/**
 * Solution for Hollow Square Pattern
 * Author: Java Tech Community by PKC
 * Approach: Using String.repeat() for edges and loops for inner rows
 */
public class HollowSquarePattern
{
    public static void main(String[] args)
    {
        int n = 5; // Size of the square
        System.out.println("* ".repeat(n));
        for (int i = 1; i <= n - 2; i++)
        {
            System.out.println("* " + "  ".repeat(n - 2) + "* ");
        }
        System.out.println("* ".repeat(n));
    }
}
```

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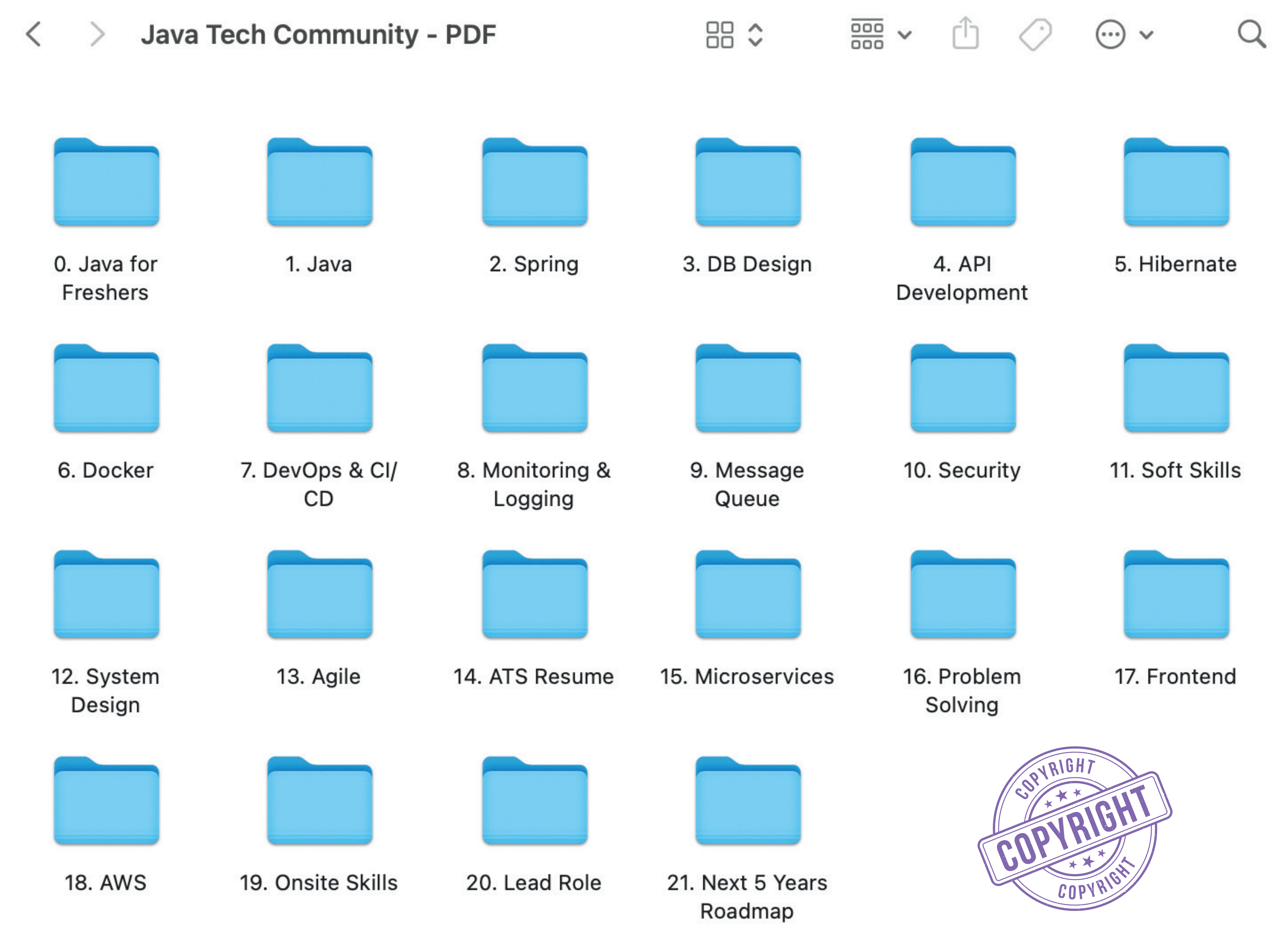

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